CIS 678 Machine Learning

Non linear Dimensionality Reduction

Dimensionality Reduction

- Linear
  - Principal Component Analysis (PCA)
  - SVD
- Non linear (Using Neural Networks)
  - Auto Encoders
  - RBMs

Non linear Dimensionality Reduction

- Auto Encoders
- Restricted Boltzmann Machines (RBMs)

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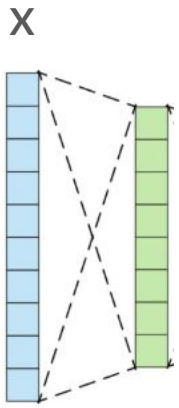

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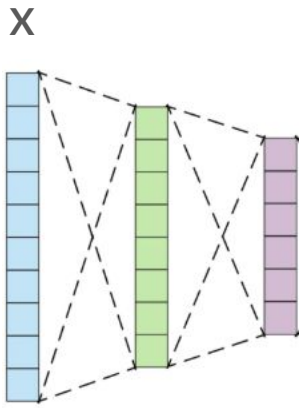

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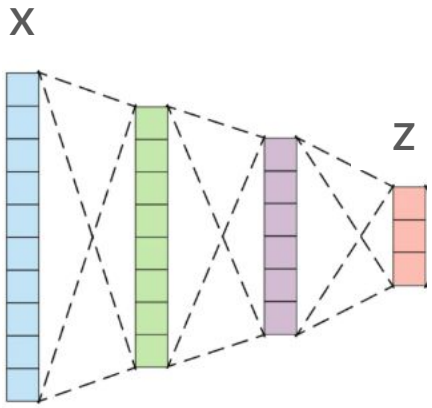

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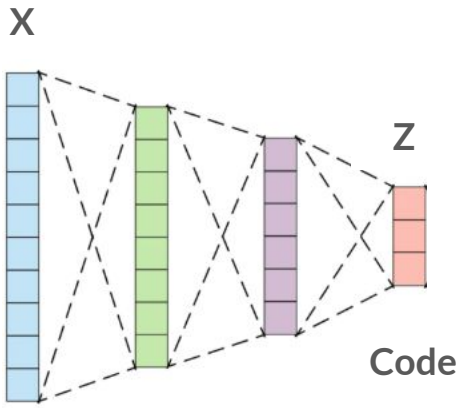

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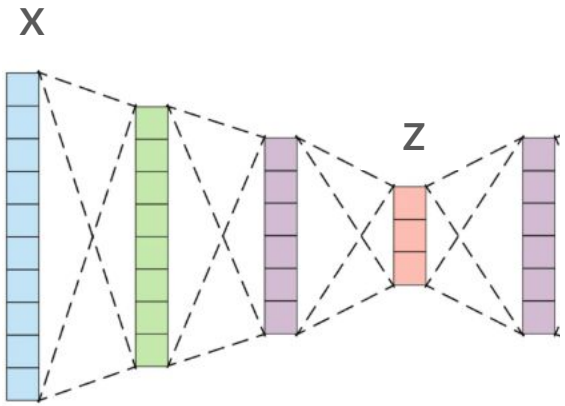

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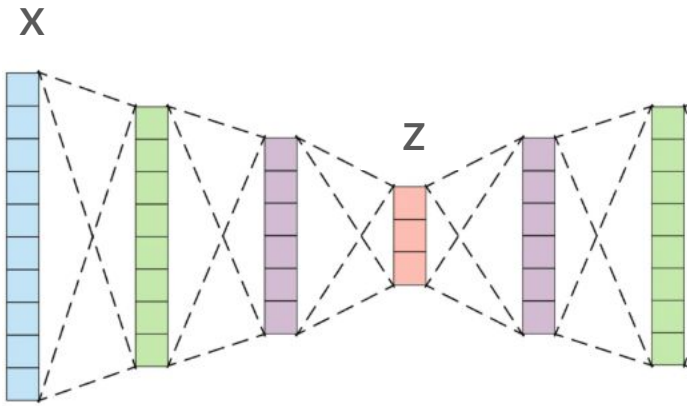

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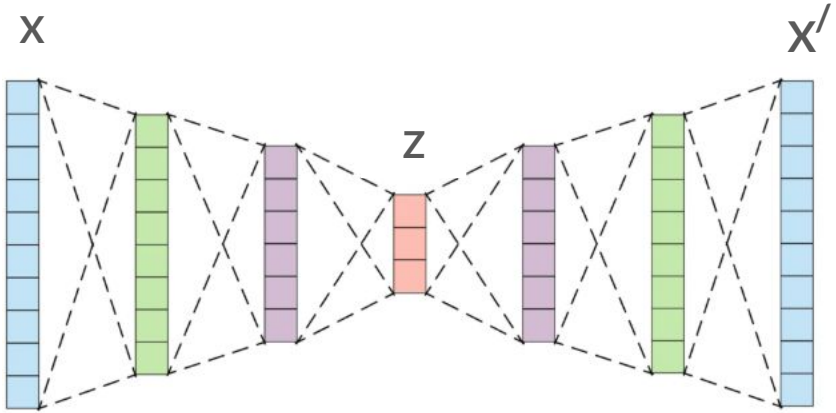

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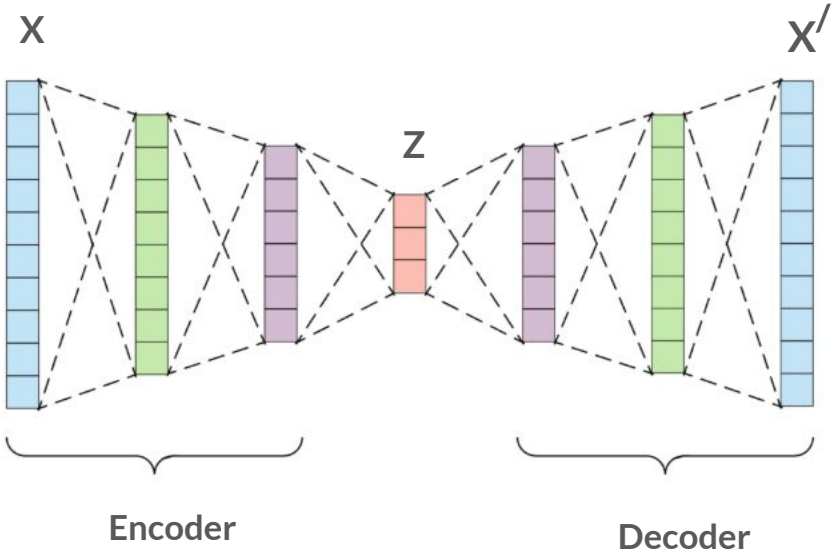

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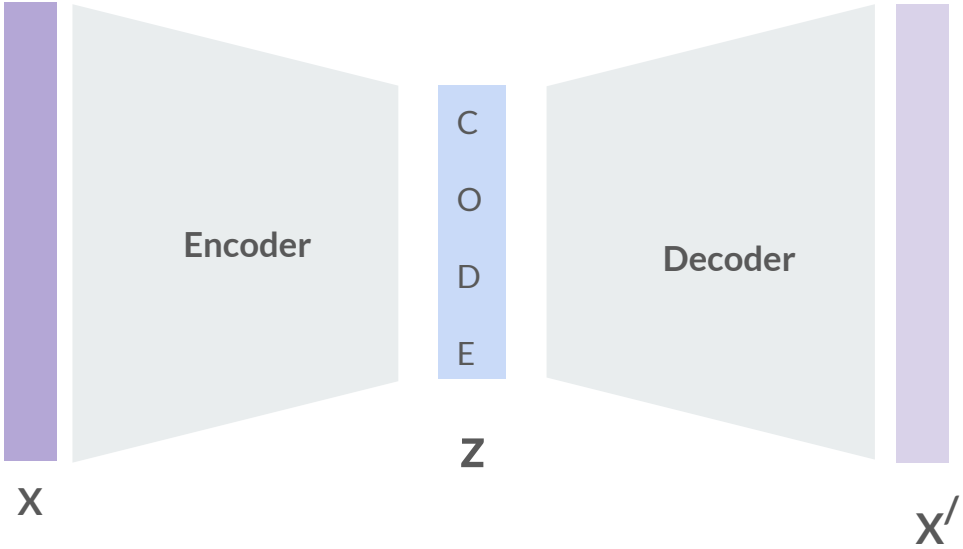
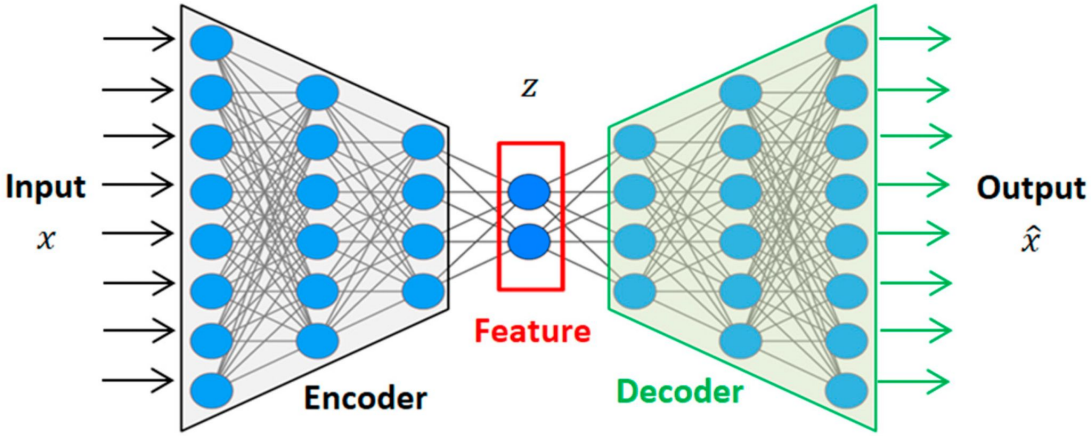
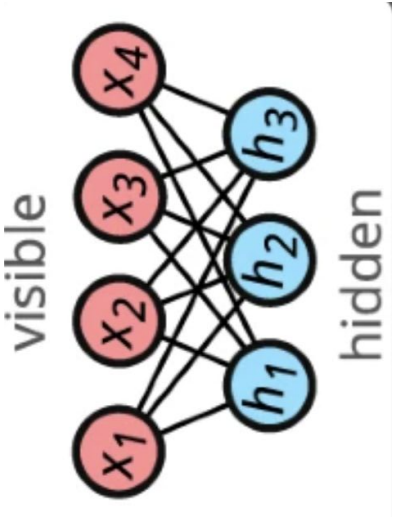


Auto encoder

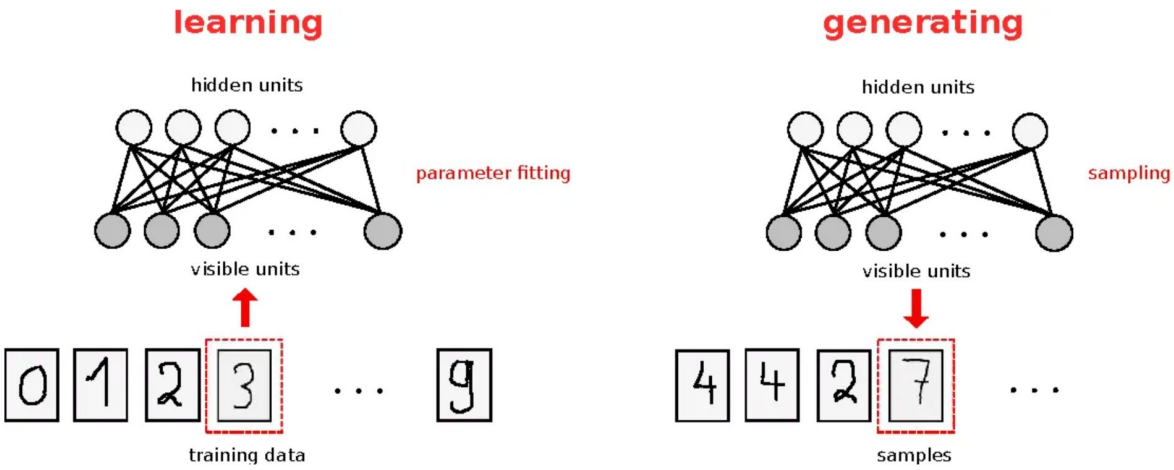
[Notebook]



Restricted Bolztman Machines (RBMs)



RBMs


learning
hidden units
parameter fitting
visible units
training data
generating
hidden units
sampling
visible units
samples

QA